GDOT Traffic Dataset

Central question: How many days of data does a partially-observed month need to be statistically representative?

Key Finding: A 14–18-day threshold may be sufficient, suggesting the current 26-day SOP is overly conservative.

2017 - 2024 | Georgia Department of Transportation

286 Stations | 8 Years | Data Understanding & Quality Review

The Data We Have: 8 Years Across 286 Stations

286 Stations Analyzed

8 Years of Data

91.6% Station-Months Complete

243 Trusted Stations

818K Clean Hourly Records (2024)

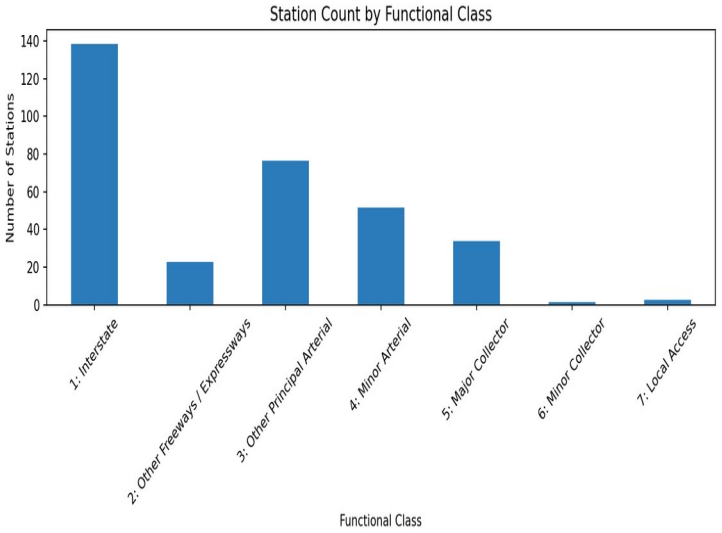


Fig. 1 — Station count by functional class

Fleet mix: 42% Interstate; ~70% urban.

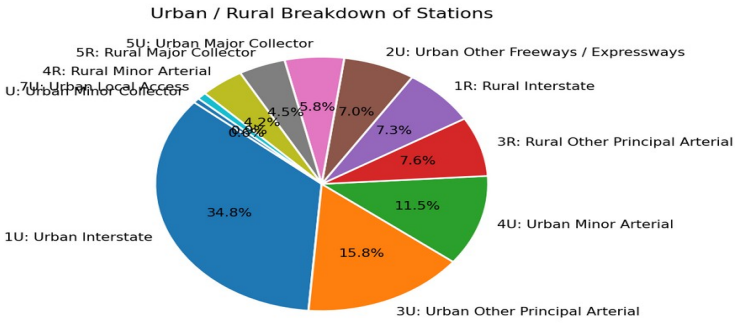


Fig. 2 — Urban/rural station breakdown

Tier complete-month rates: 93.3% / 94.9% / 98.8%; mean station = 72.9 complete months.

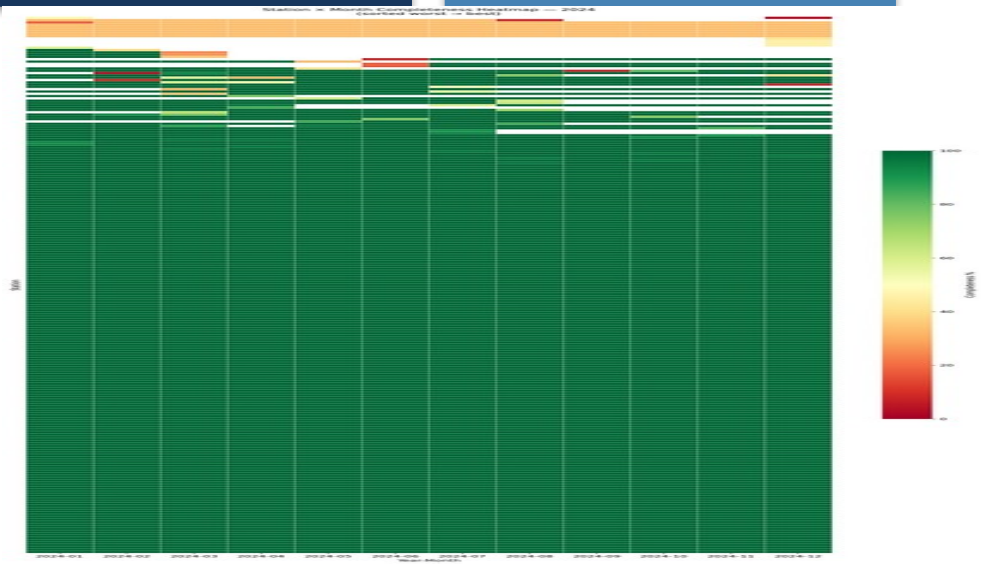


Fig. 3 — Station-month completeness heatmap

Heatmap: mostly complete station-months; white cells mark gaps.

91.6% Complete — But the Missing 8.4% Has a Pattern

91.6%	8.4%	73.4%	1.8%	226
Data Complete	Overall Missingness	Always-On Stations	Chronically Sparse	Outage Runs (max 60 mo.)

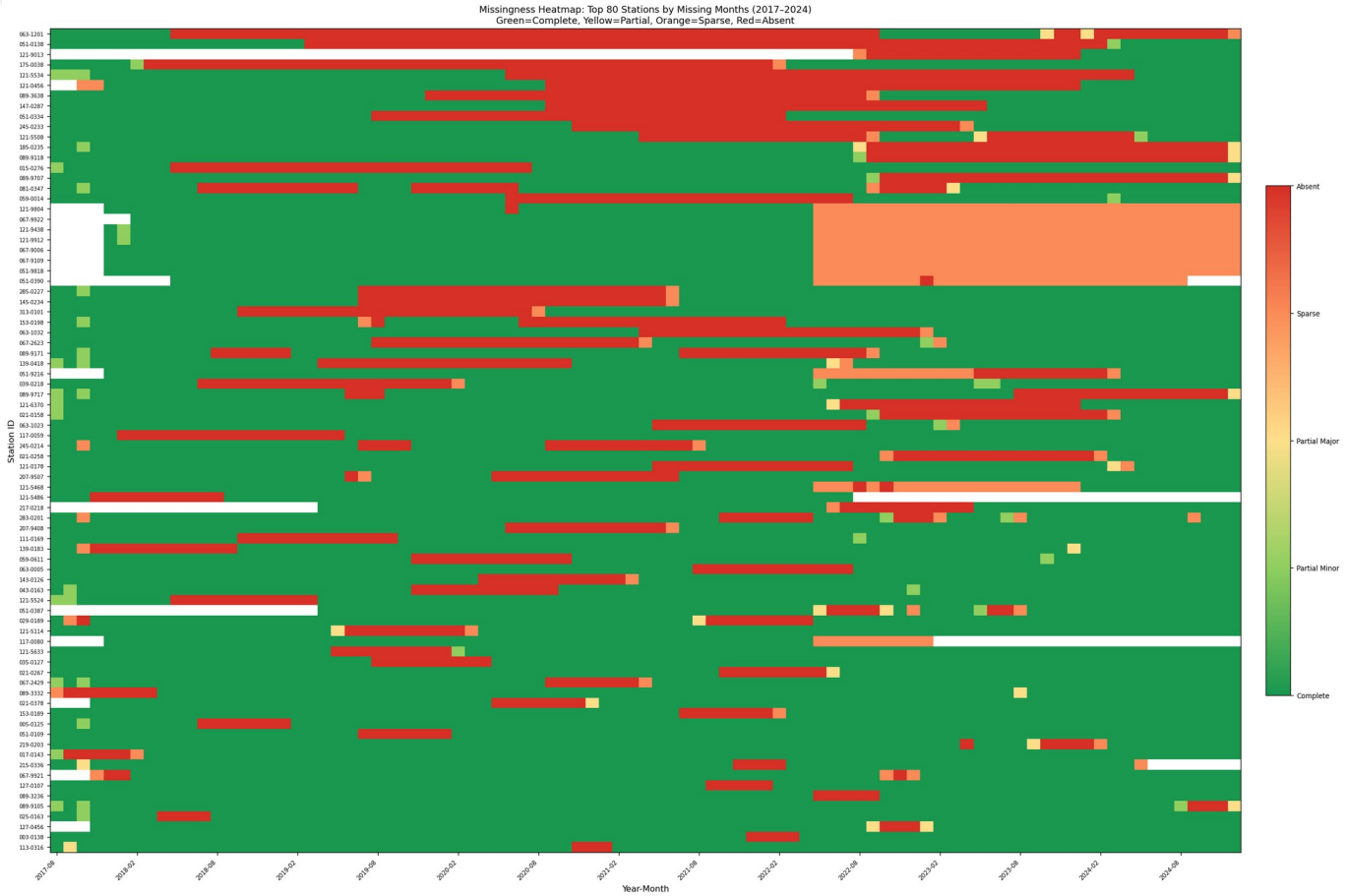


Fig. 5 — Station × date missingness heatmap (each row = one station, each column = one month, 2017–2024)

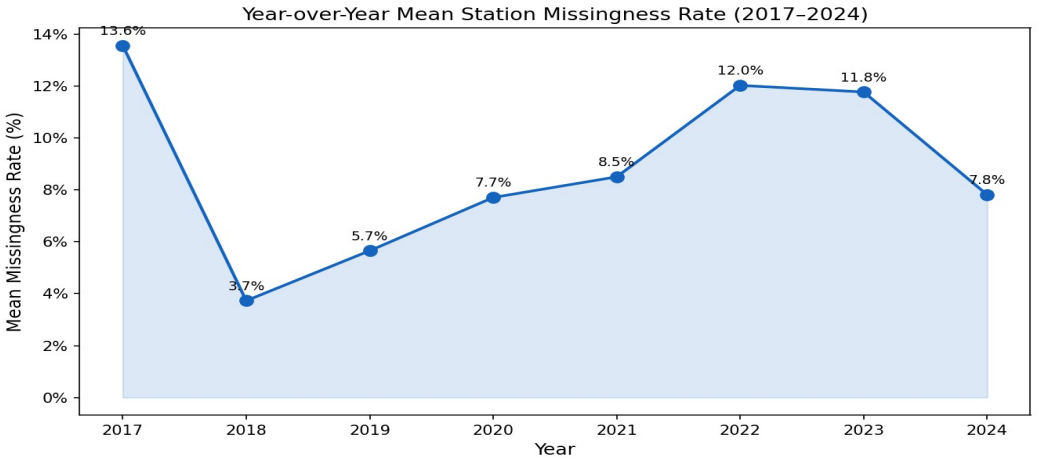


Fig. 6 — Annual missingness rate (2017–2024); spikes in 2017 and 2022–23 reflect deployment and pipeline transition events

2017: 13.6%	Incomplete initial deployment — startup artifact
2022–23: 12.0%	Spike from pipeline transition to 5-min system
2024: 7.8%	Recovering, but not yet back to baseline
Pattern:	Gaps cluster around operational events, NOT random noise

Modeling Implication: Missingness follows a non-random, temporally structured pattern — gaps spike during known deployment and system transitions, not uniformly across time.

The Interstate "Missingness" May Be a Counting Artifact — Not a Data Failure

MISSINGNESS BY FUNCTIONAL CLASS

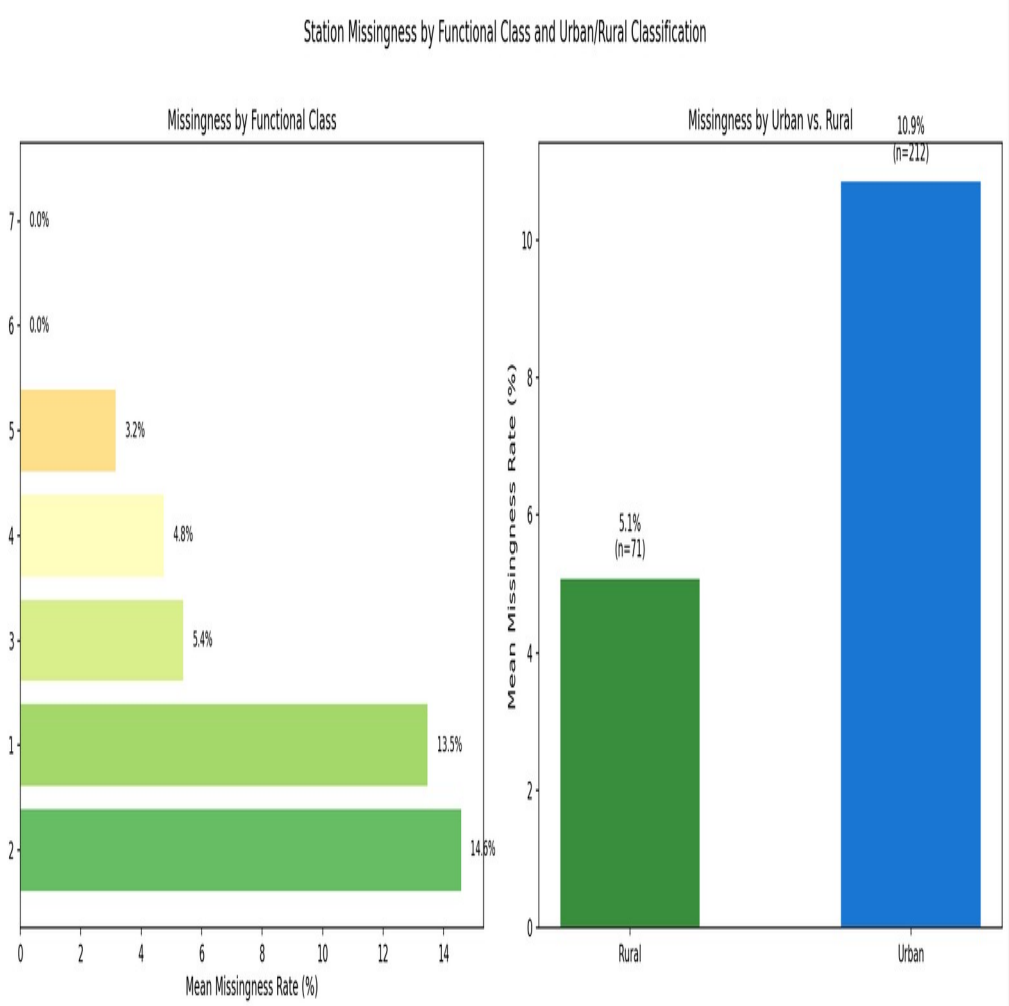


Fig. 7 — Missingness rate by functional class; FC1/FC2 elevated due to multi-lane sensor counting methodology

FC MISSINGNESS RATES

FC1 Interstate	13.5%
FC2 Freeway	14.6%
FC3 OPA	5.4%
FC4 Arterial	4.8%
FC5 Collector	3.2%

WHY?

Multi-lane sensors generate more expected readings per interval, making the 80% completeness threshold harder to meet — not necessarily worse data quality.

52% of outage runs are 1 month (maintenance). 32% exceed 6 months (sensor replacement).

OUTAGE RUN-LENGTH DISTRIBUTION

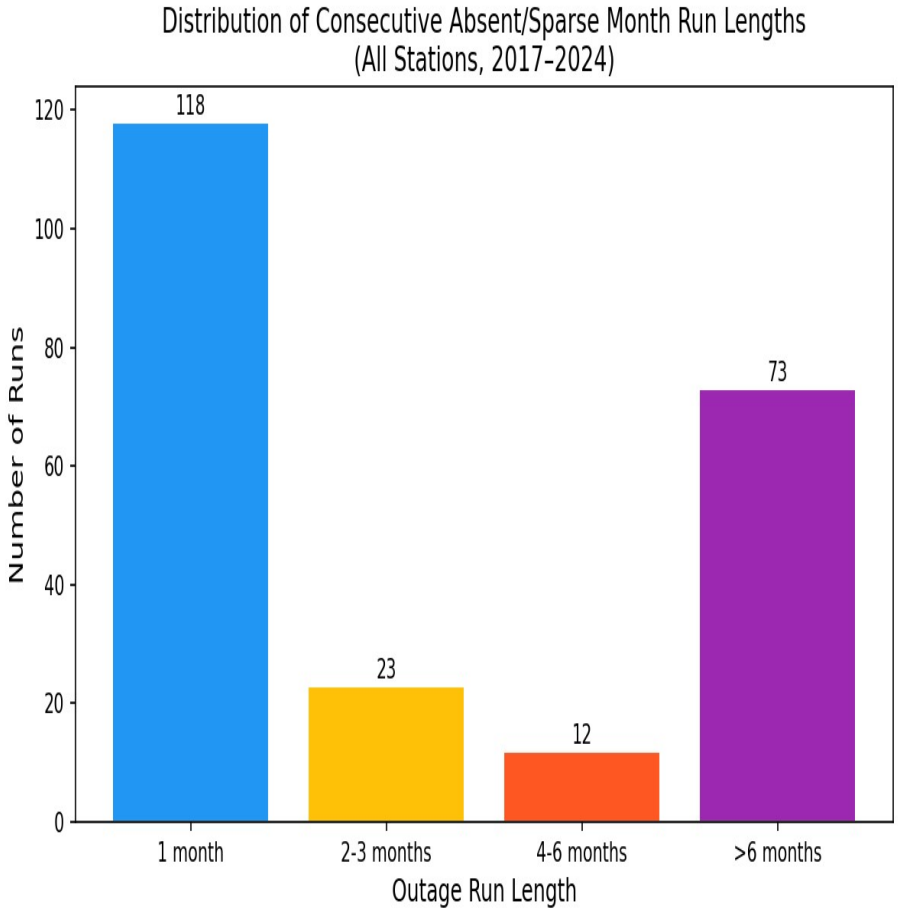


Fig. 8 — Run-length distribution of consecutive missing months; 52% are single-month outages

Research Finding: Interstate Block(1) outage patterns → $k^* = 14$ days. The 26-day SOP may discard certifiable months. Validate threshold before excluding FC1/FC2 stations.

The 26-Day SOP Fits No Station Perfectly — k* Depends on Traffic Volume

MINIMUM DAYS (k*) BY TRAFFIC VOLUME TIER — GDoT paper, 84 stations, April 2024

Volume Tier & Road Type	k*	Network	vs. 26-day SOP
High-volume >15,000 MADT *FC1/FC2 — Interstates & Freeways*	**14**	42%	Over-restricts by 12 days
Medium-volume 5,000–15,000 MADT *FC3/FC4 — Arterials*	**18**	~50%	Over-restricts by ~8 days
Low-volume <5,000 MADT *FC5 — Major Collectors*	**24**	~8%	SOP ≈ right; some stations never reach sufficiency at 28d

Source: GDoT paper (Georgia Tech, 2026) — network-wide simulation, 84 eligible stations, April 2024, linear estimator, α = 10%.

KEY FINDING

Traffic volume is the primary driver of k*, not missingness type. High-volume interstates need as few as 14 days; low-volume collectors need 24+. A flat 26-day rule is too conservative for the high-volume majority (42% of the network) and may still be insufficient for some low-volume stations.

PHASE 2

Apply volume-tier-aware conformal thresholds station-by-station using the 27 identified reference station candidates.

STATION-MONTH COVERAGE DISTRIBUTION (k*=18 midpoint threshold)

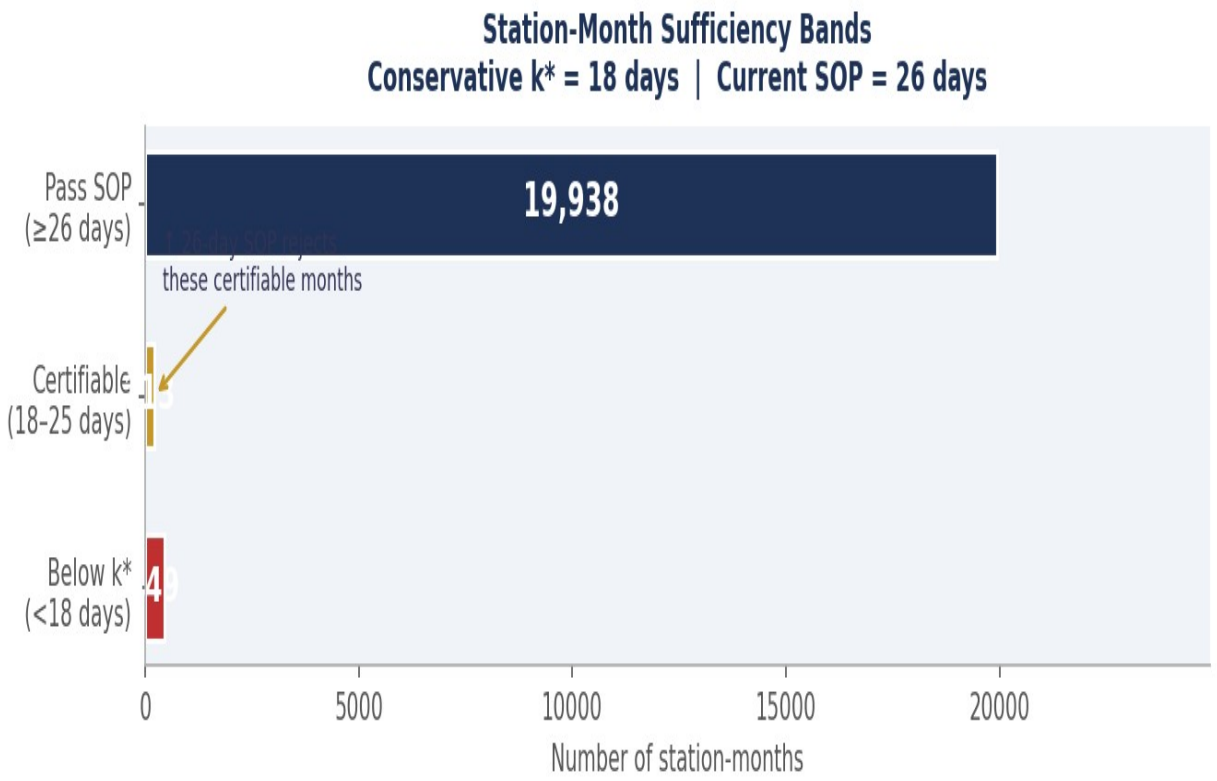


Fig. 10 — Coverage bands using k=18 as an illustrative midpoint (valid for medium-volume stations). High-volume interstates: certifiable at k*=14; low-volume collectors: require k*=24.*

96.8% of station-months already exceed 26 days (pass SOP). The 1.0% certifiable band (18–25 days, ~213 station-months) is recoverable for medium-to-high-volume stations. The 2.2% below 18 days are genuinely insufficient.

Research Finding: The 26-day SOP over-restricts high-volume stations (k* = 14) and is approximately right for low-volume collectors (k* = 24). A volume-tier-aware threshold — not a single cutoff — is the correct instrument.

Can We Trust This Data? Mostly, Yes

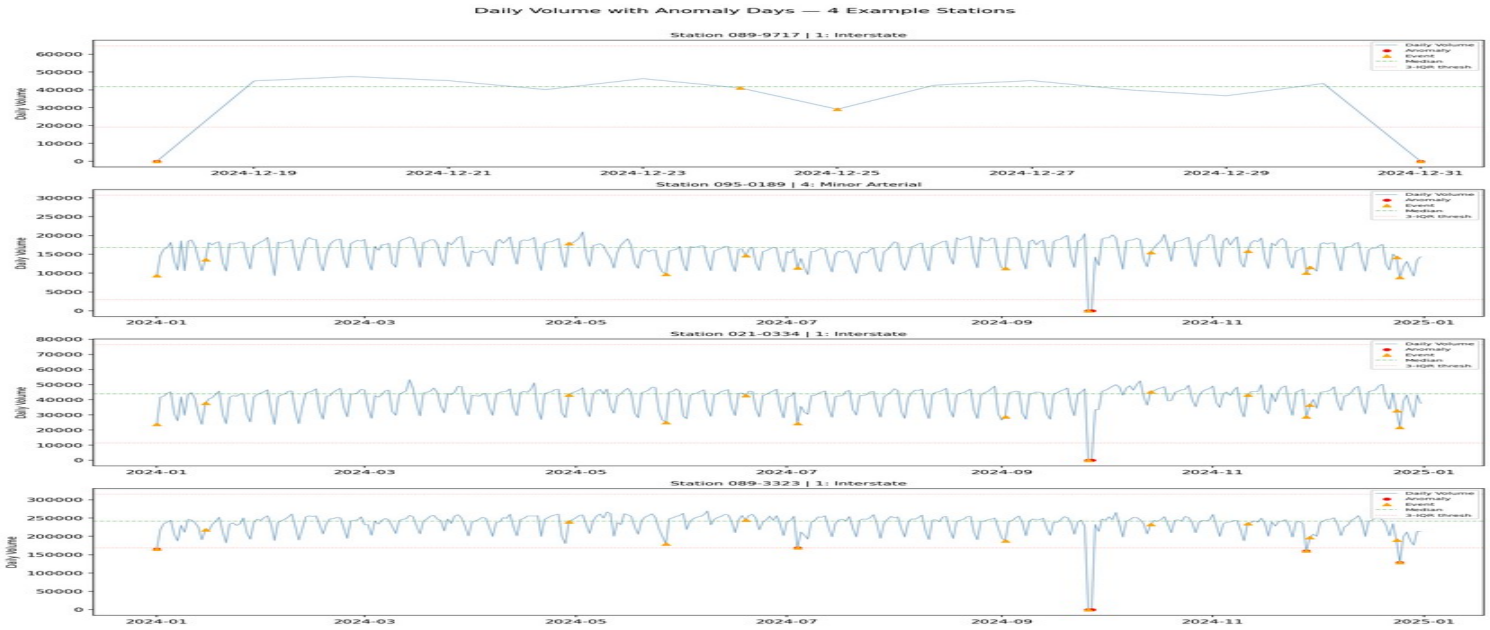


Fig. 11 — Anomaly-day counts by station (rolling MAD + IQR method, 2024 sample)

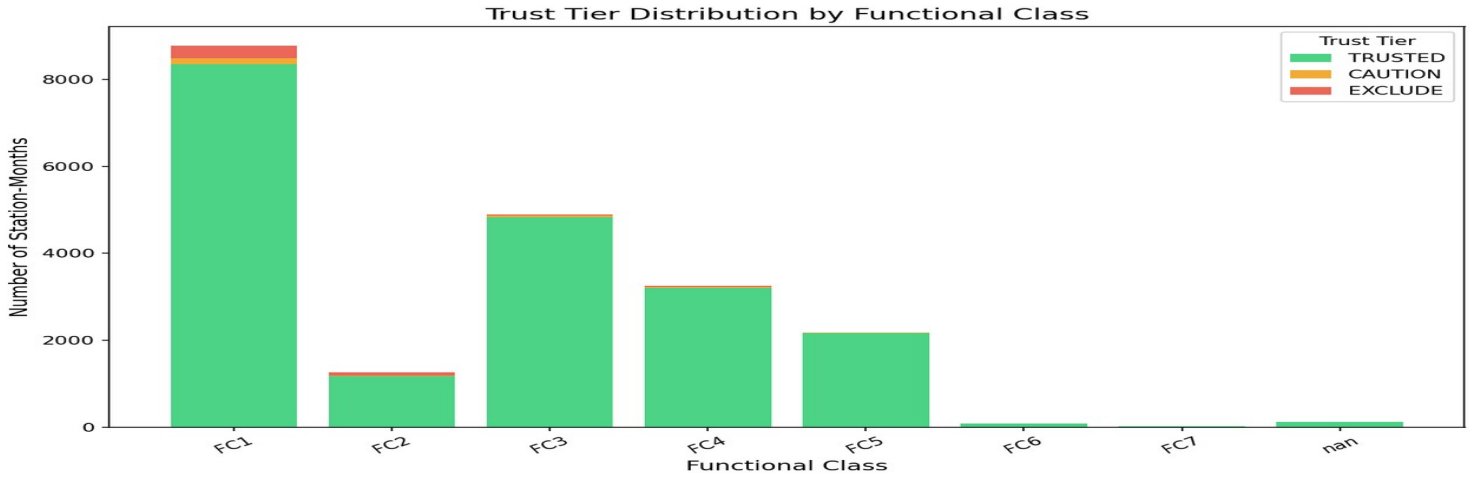


Fig. 12 — Station trust tier summary: TRUSTED / CAUTION / EXCLUDED

ANSWER:

The data is trustworthy.

84.9% of stations pass all quality checks.
90.5% of detected anomalies are explained by known events in the GDOT event database.

Anomaly Detection (58 stations, 2024):

391 anomaly-days found. 354 explained (90.5%). 37 unexplained (9.5%).

Station Trust Tiers (286 Stations)

Tier	Station-Months	Stations
TRUSTED	19,963 (96.9%)	243 (84.9%)
CAUTION	233 (1.1%)	36 (12.6%)
EXCLUDED	404 (2.0%)	7 (2.5%)

What this means:

243 TRUSTED stations form the primary modeling population. 36 CAUTION stations can be used in sensitivity tests. 7 EXCLUDED need manual investigation.

Traffic Is Predictable: Same Peaks, Every Day, Every Season

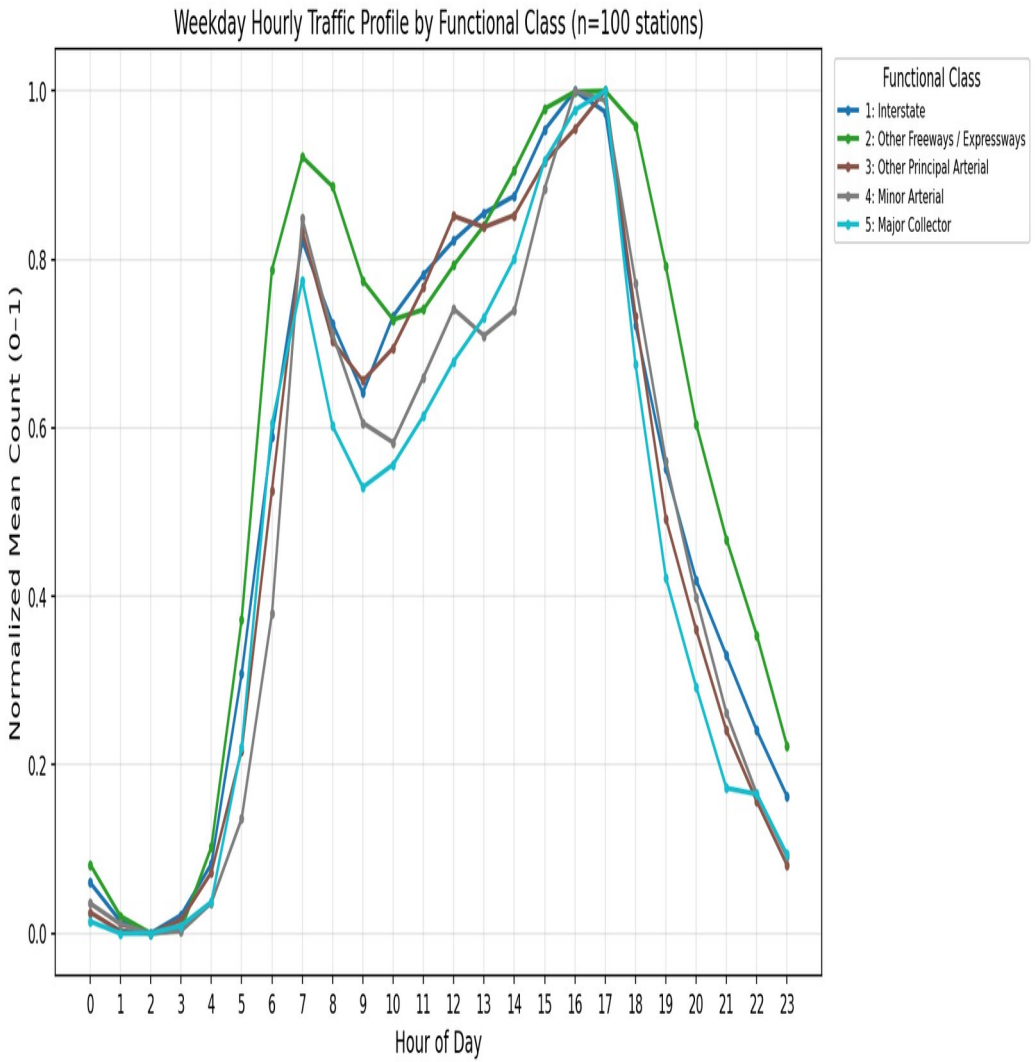


Fig. 13 — Mean hourly volume profile (100 stations, 2024): AM peak 07:00, PM peak 17:00, consistent all seasons

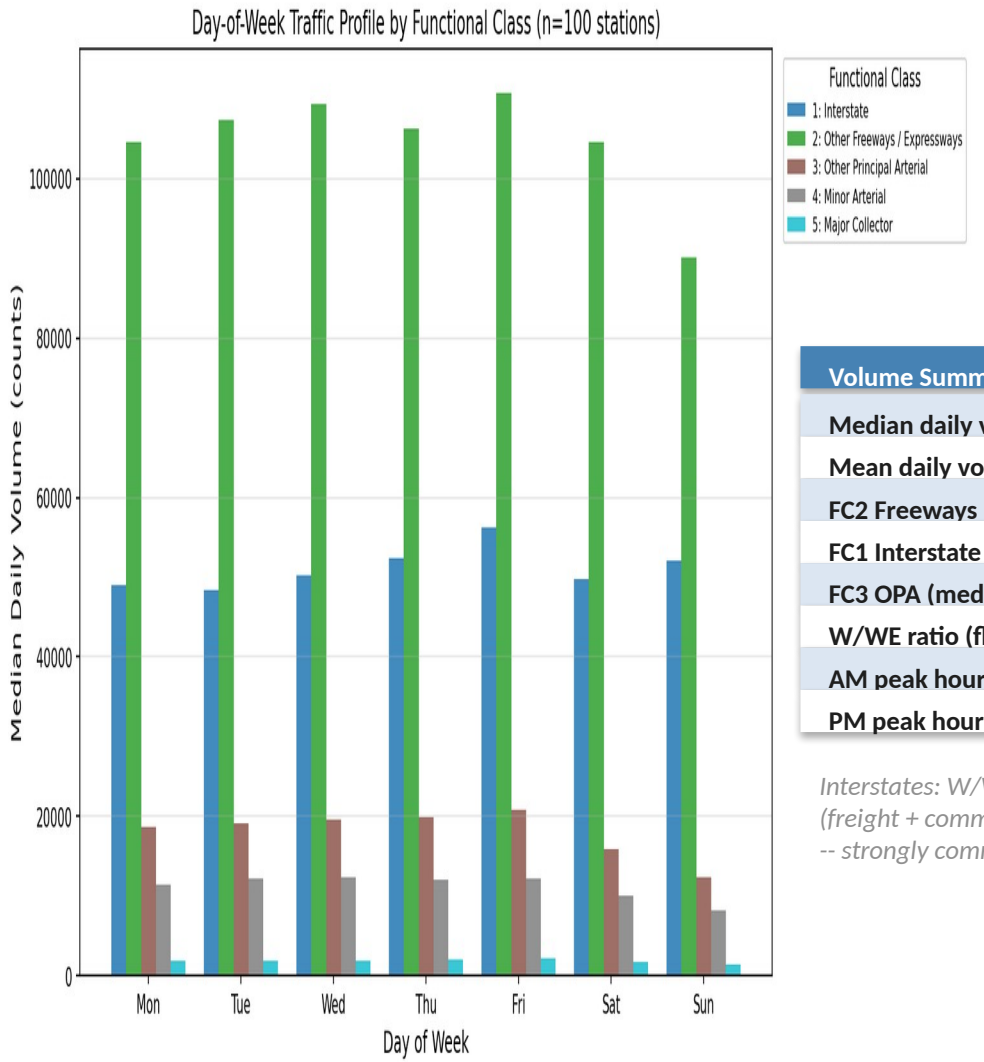


Fig. 14 — Day-of-week volume pattern: weekday-to-weekend ratio = 1.25

Volume Summary (100 stations, 2024)	
Median daily volume	22,830 veh/day
Mean daily volume	45,223 veh/day
FC2 Freeways (median)	~110,000 veh/day
FC1 Interstate (median)	~52,000 veh/day
FC3 OPA (median)	~19,000 veh/day
W/WE ratio (fleet avg)	1.25 -- weekday-
AM peak hour	07:00 (stable all
PM peak hour	17:00 (stable all seasons)

Interstates: W/WE ratio 1.08 -- nearly uniform 7-day use (freight + commuter mix). Arterials/Collectors: 1.35-1.44 -- strongly commuter-oriented.

Why This Matters: The AM/PM peaks sit at 7:00 and 17:00 across every road type and all four seasons. This consistency is what makes the dataset easy to reason about -- daily traffic behaves in a stable, repeatable way.

COVID Dropped Traffic 8.6%. It Came Back by 2022.

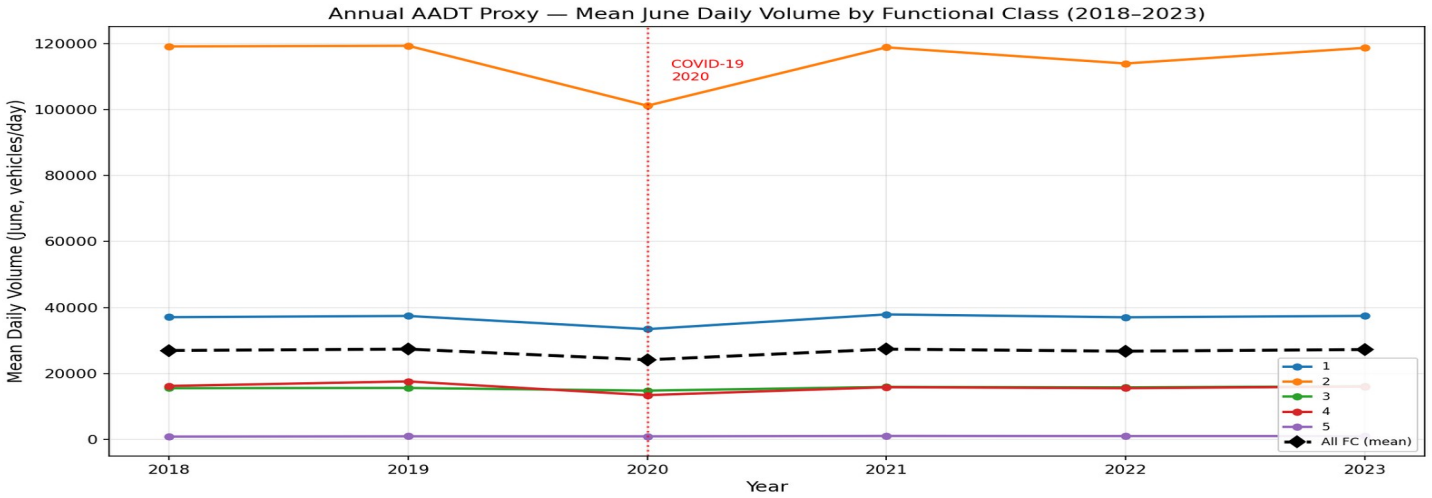


Fig. 16 — AADT proxy trend (30-station sample, 2018–2023): COVID-19 drop in 2020, recovery by 2022

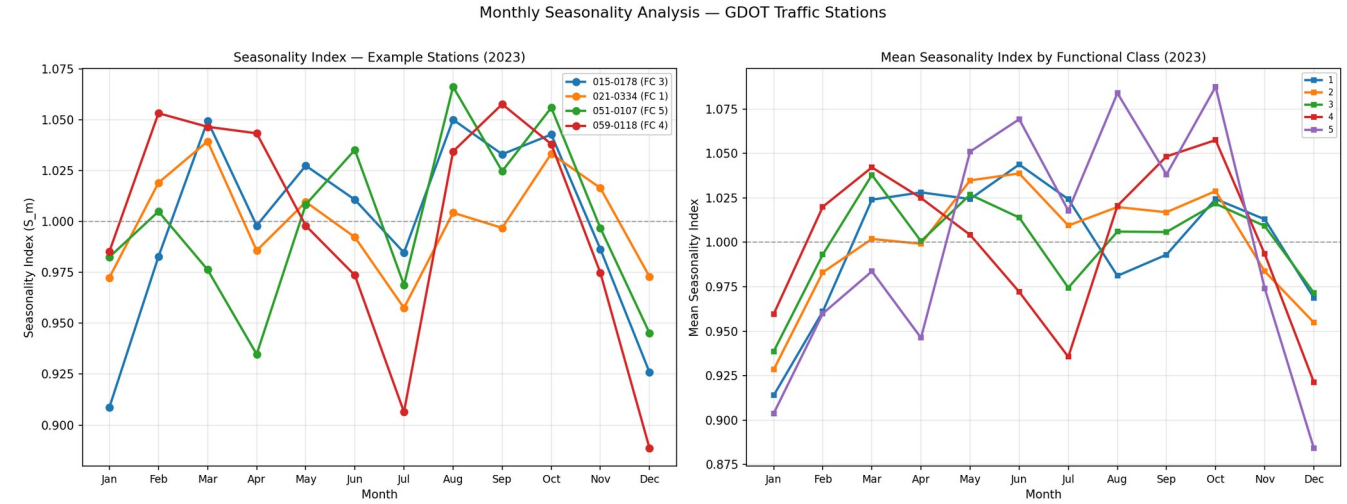


Fig. 17 — Seasonal index by month (2023 baseline): traffic peaks in October and bottoms in January; annual swing ≈ ±10%

AADT Proxy Trends (30-station sample, 2018-2023)

COVID-19: median volume dropped ~8.6% from 2019 to 2020 across all FC

Recovery: near-2019 levels restored by 2021-2022 for most stations

Freeways recovered fastest; major collectors were slowest

AADT proxy = June mean daily volume -- least seasonally biased month; avoids requirement for a full calendar year of data

Seasonality (2023 baseline)

Peak month: October (Seasonal Index = 1.036)

Trough month: January (SI = 0.928) -- +/-10% annual amplitude

Rural FC4-5 stations show stronger seasonality than Interstate

Peak hours (7 AM / 5 PM) are stable across all 4 seasons

Why This Matters:

Two forces shape the long-run trend: a real but temporary COVID dip (fully recovered by 2022) and a regular seasonal cycle -- traffic peaks in October and bottoms in January, a roughly +/-10% swing that is stronger on rural roads. Any reading of the data has to account for both.

Holiday Effects Are a Bias Risk — First Fix the ID Join

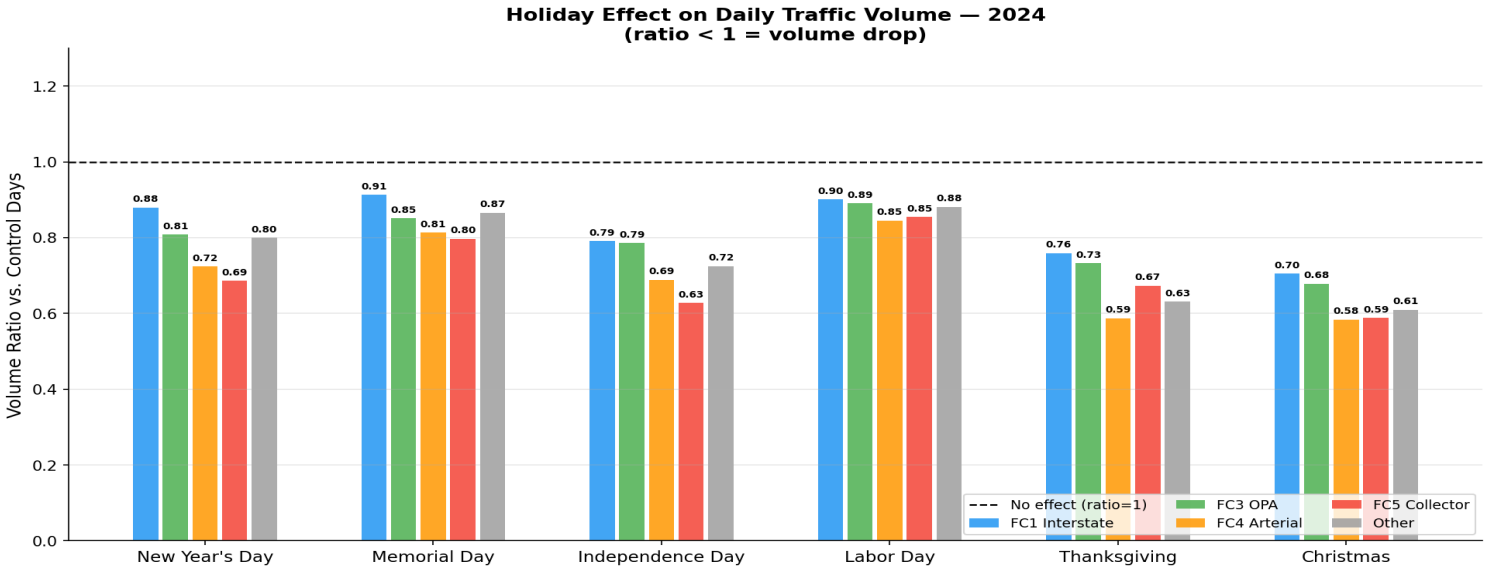


Fig. 18 — Holiday-effect diagnostic: current output is empty/NaN due to station-ID mismatch

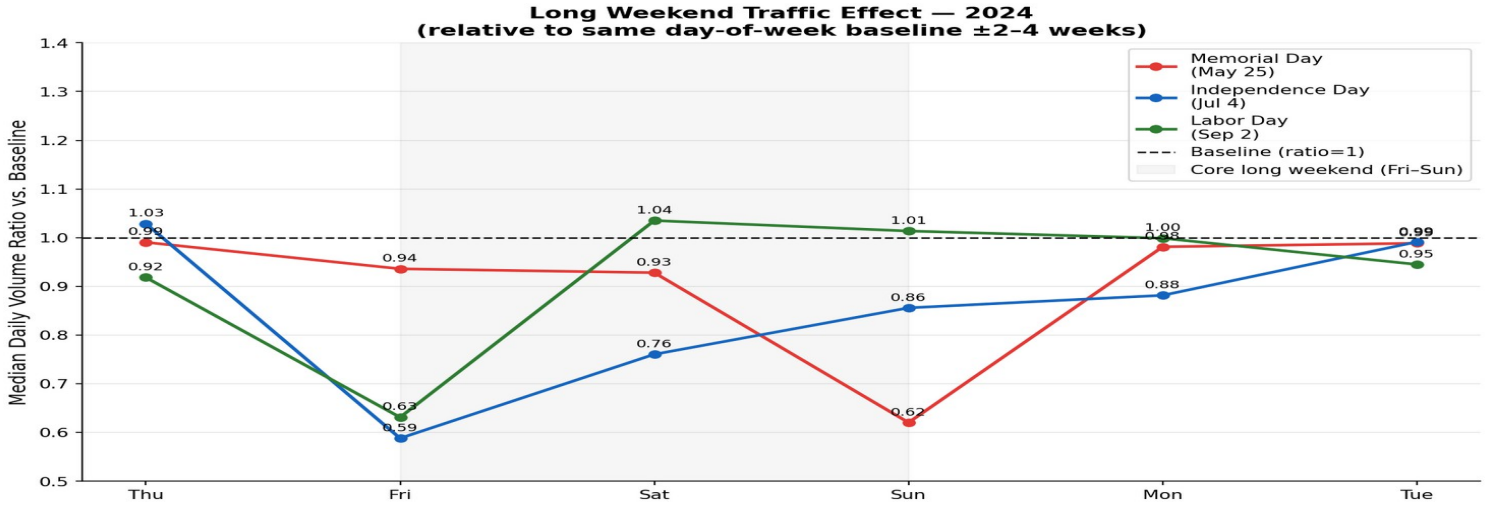


Fig. 19 — Long-weekend diagnostic: same ID-matching issue blocks interpretation

What the Graphs Show

Holiday and long-weekend checks were attempted.
Both outputs do not produce usable ratios yet.

- The issue appears systematic, not holiday-specific.

Main Concern

Event station IDs do not match 2024 COSIT traffic files.
Unreliable joins can create misleading correction factors.

- Holiday/long-weekend traffic may still bias monthly AADT.

Nearby Stations Tend to Behave Alike

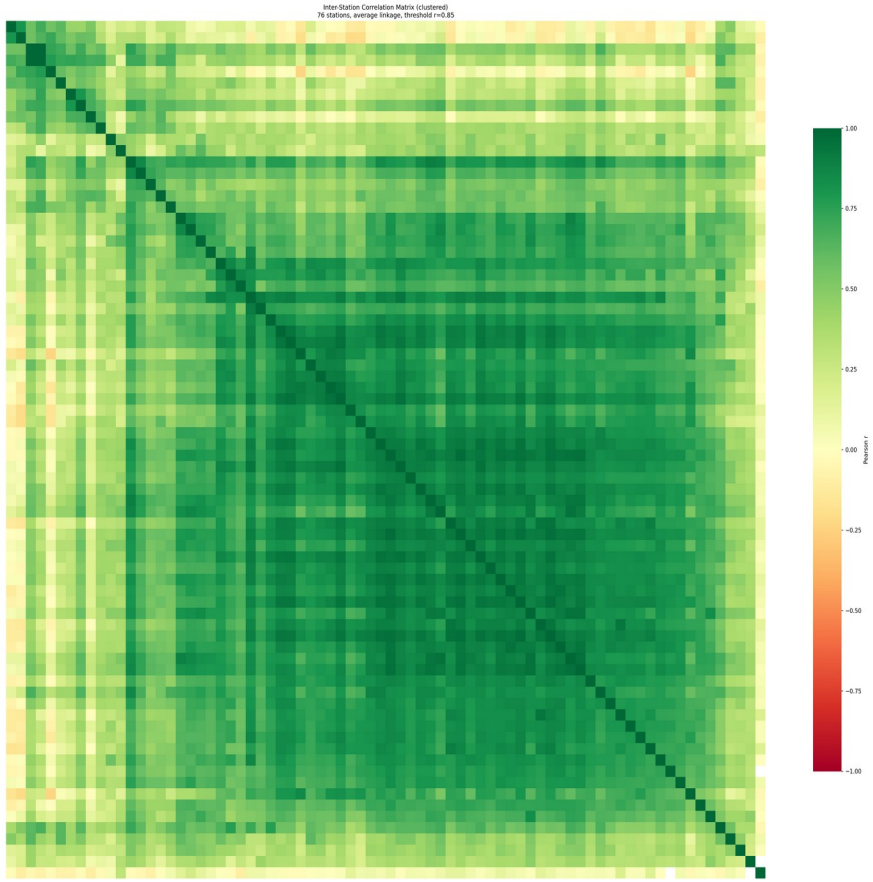


Fig. 20 — Pairwise Pearson correlation matrix (top 80 stations, ≥180 overlapping days); 411 high-correlation pairs (r > 0.85)

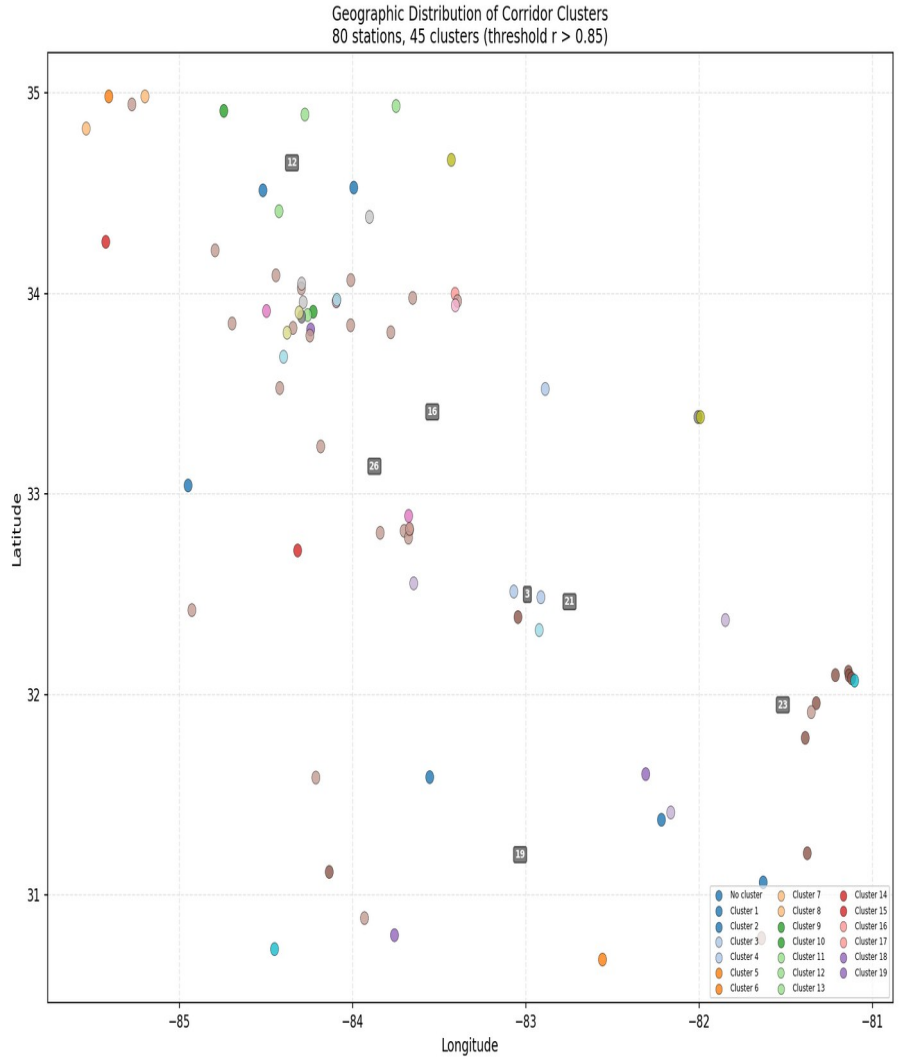


Fig. 21 — Behavioral neighbor pair: daily volumes on same corridor (r ≥ 0.95 typical for corridor neighbors)

Pairwise Correlation (top 80 stations)

Valid pairs	2,848 (>=180 overlapping
Mean Pearson r	0.573 (median 0.636)
High-corr (r>0.85)	411 pairs (14.4% of all pairs)

Clusters and Reference Stations

- 7 multi-station corridor clusters identified
- 211 behavioral neighbor pairs (r>0.7, dist<50km)
- 27 reference stations pass all 5 quality criteria
- Top: 089-3323 (DeKalb, Interstate, 10 neighbors)

Note: analysis covers top 80 only -- ~150 more eligible stations not yet assessed

Why This Matters: Stations on the same corridor rise and fall together. The 27 reference candidates give a practical donor pool for gap-filling.

What the EDA Produced: A Clean, Characterized Dataset

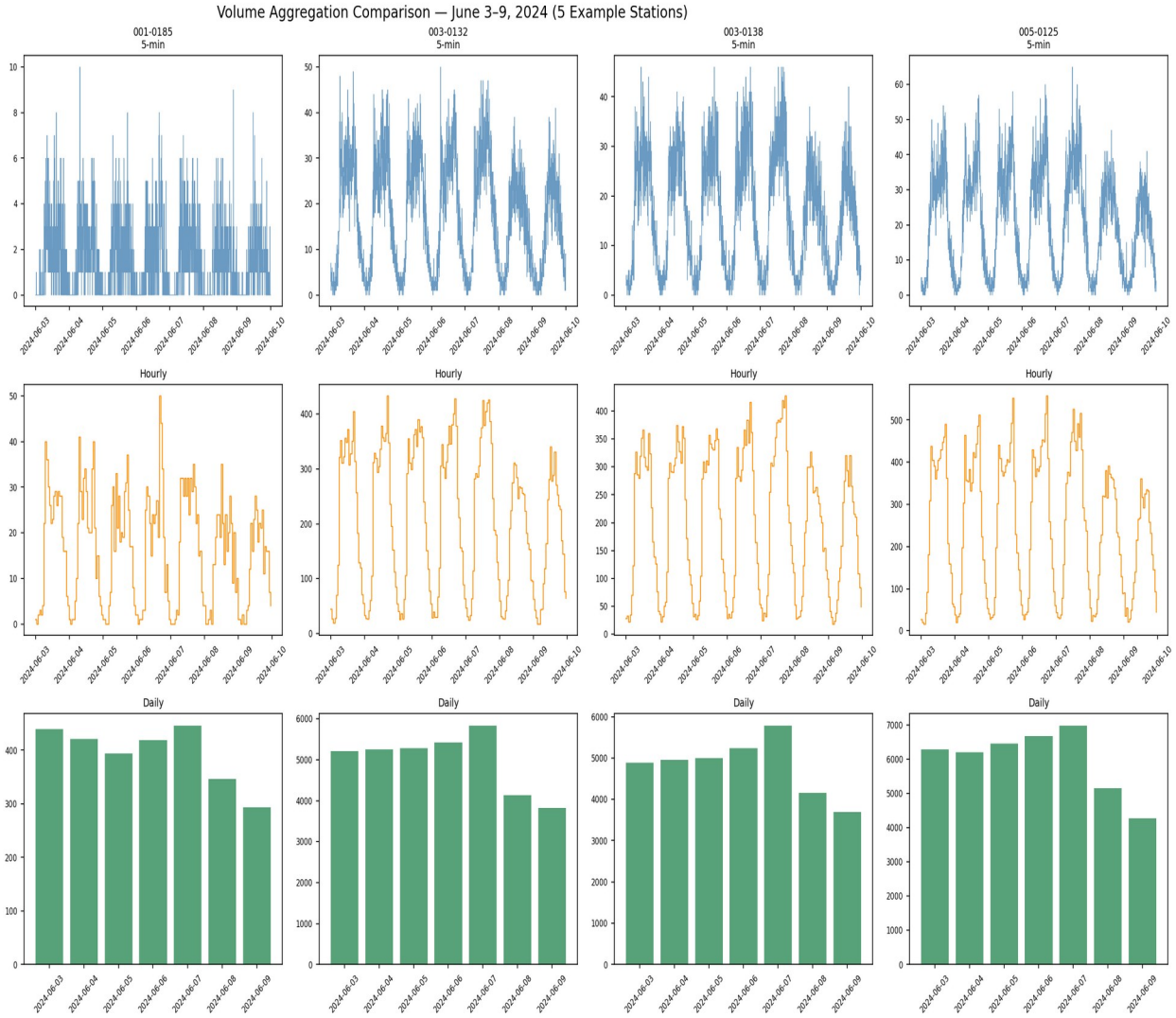


Fig. 22 — Clean hourly subset and engineered feature table overview (818K records; 99 processed stations; 286-station feature table)

Clean Hourly 2024 Dataset	
Hourly records	818,645
Trusted stations covered	99 of 207 eligible
Temporal span	Full calendar year 2024
Mean hourly coverage	100% (min threshold: 50%)
Imputed hours	1.7% (flagged, not removed)
Median hourly volume	580 veh/hr (max 17,822)

Station Feature Table	
Dimensions	286 stations x 27 features
AADT proxy	median daily x 365 (June-reference-free)
Seasonal amplitude	per station from 2023 data
Recommended resolution	Hourly -- preserves AM/PM peaks, reduces noise

Datasets and Reference Files Produced	
station_feature_table.csv	286 stations x 27 modeling features
clean_hourly_2024_trusted.csv	818K hourly records, 99 stations, 26 cols
reference_station_candidates.csv	27 top gap-filling donor stations
trusted_station_list.csv	243 HIGH-trust stations
holiday/event ratios	pending COSIT lookup fix

AI-Assisted Subagent Pipeline Design

8 specialized agents decomposed the EDA into independent, auditable analysis tracks. All outputs were human-reviewed before being accepted.

PIPELINE FLOW

1 station_inventory

Counted all 286 stations, years, months. Audited schema across 3 tiers. Flagged COSIT ID format mismatch.

2 missingness_quality

Computed % missing by station/hour/DOW. Labeled outage patterns. 91.6% overall completeness confirmed.

3 traffic_profile

Built hourly, DOW, and monthly profiles. Confirmed AM peak 7:00, PM peak 17:00 stable across all seasons.

4 temporal_trend

Tracked AADT proxy trends and seasonality. COVID drop 8.6% in 2020, recovered by 2022. Holiday ratios remain blocked by ID mismatch.

5 corridor_relationship

Pairwise correlations for the top 80 complete 2024 stations. Identified 27 reference candidates for gap-filling.

6 anomaly_representativeness

Rolling MAD + IQR anomaly flags. Assigned trust tiers: 243 TRUSTED / 36 CAUTION / 7 EXCLUDED.

7 feature_pipeline

Produced 818,645 clean hourly records for 99 trusted stations. Built a 286-station, 27-feature table.

8 eda_manager

Synthesized agent outputs into the master EDA summary, roadmap, AI usage report, and presentation deck.

8	286	818K	243	27
Specialized Agents	Stations Analyzed	Clean Records	Trusted Stations	Reference Candidates

Key Takeaways

1 The Data Is Trustworthy

84.9% of stations pass all quality checks. 96.9% of 20,600 station-months are TRUSTED. Only 7 stations (2.5%) are excluded — a strong foundation for modeling.

2 Missingness Is Structured, Not Random

91.6% of expected records are present. Gaps cluster around known operational events (2017 startup, 2022–23 pipeline transition) — not random sensor noise. Behavioral-neighbor gap-filling is the correct response.

3 Traffic Patterns Are Stable and Interpretable

AM peak 07:00, PM peak 17:00. COVID-19 dropped traffic 8.6% in 2020, with recovery visible by 2021-22. Holiday correction factors are not yet quantified.

4 14–18 Days Is Enough for Monthly Certification

The current 26-day SOP is too conservative. All observed outage patterns (MCAR, Block, Bursty) map to $k^* \leq 18$ days via nearest-neighbor conformal testing — saving ≥8 days of discarded data per partial month.

5 Corridor Neighbors Support Gap-Filling

Top-80 correlation analysis found 211 behavioral-neighbor pairs and 27 reference candidates. Corridor donors are a stronger gap-filling starting point than fleet-wide averages.

6 Modeling Is Close, With Clear Remaining Blockers

The 818K hourly records cover 99 trusted stations. Before full modeling, remove the 100-station cap, fix holiday COSIT matching, expand corridor analysis, and avoid current archetype labels.

Research Questions from Reading the GDoT Report

3 Weighted Conformal Prediction

Proposed improvement: weight reference months by similarity instead of treating them equally

Proposed weight formula:

$$w_i = \exp(-\lambda (1 - r_i))$$

where r_i = Pearson correlation between reference month i and the target month

Why this helps:

- Keeps more reference months without sacrificing certification quality
- Reduces influence of weakly-correlated months on the test statistic
- Amplifies the contribution of the most similar reference months
- Improves coverage for rural stations with small reference pools

The gap this addresses:

Only 84 of 185 eligible stations passed the pool-size requirement. The failed stations were mainly low-volume rural stations whose traffic profiles were too variable to find strong corridor neighbors. A weighted scheme may certify more of them without relaxing the exchangeability guarantee.

4 Reference Pool Sensitivity

How sensitive is conformal prediction to how we construct the reference pool?

"Since conformal prediction depends on the reference months being comparable to the target month, how sensitive is the method to how we construct the reference pool?"

Current method:

Accept months above a fixed Pearson correlation threshold; treat all accepted months with equal weight.

The trade-off:

Relax threshold →	More reference months, but weaker exchangeability guarantee
Tighten threshold →	Fewer months, stronger guarantee — but may fail rural stations

Follow-up question:

"For example, if we relax the threshold we get more reference months, but maybe weaker exchangeability. How should we balance reference pool size versus similarity quality?"

Potential direction:

Combine ideas 3 & 4: a weighted pool whose effective size adapts based on correlation drop-off, rather than a fixed binary threshold.